

Induced Representation

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Problem 1.

Proof. $\forall f \in \text{Hom}(G, U)$, $\forall g_1, g_2 \in G$, $(g_1 g_2 f)(x) = f(x g_1 g_2) = (g_2 f)(x g_1) = g_1(g_2 f)(x)$. Thus the action is a valid group action.

J is the subset of functions of $\text{Hom}(G, U)$ which satisfy $F(hx) = hF(x)$.

$\forall g \in G$, $\forall F \in J$, $\forall x \in G$ and $\forall h \in H$, $(gF)(hx) = F(hxg) = hF(xg)$. Thus $gF \in J$. Thus $gJ \leq J$, thus J is preserved by G action.

$\forall Hx \in H \setminus G$, $\forall hx \in Hx$, $\forall F \in J$, $F(hx) = hF(x)$. Thus $F(x)$ determines the coset. Let $J_x \leq J$ be the set of functions which map $g \in G \ni g \notin Hx$ to 0. $F(x)$ is equal to some $J(x)$ for some $J_x(x) \in J$. There is a map from $J_x \rightarrow U$, where $f \rightarrow f(x)$. The map is a homomorphism, because $f_1 f_2(x) = f_1(x) f_2(x)$. The map is surjective because $\forall u \in U \exists f \in J_x, f(x) = u$. Moreover everything in the kernel must map x to 0 and thus everything in the coset to 0 and thus must be the trivial map. Thus the map is an isomorphism, thus $J_x \cong U$. There are $|H \setminus G|$ cosets each with a corresponding J_x and $\forall k \in J_{x_1} \cap J_{x_2}$, where $x_1, x_2 \in H \setminus G \ni x_1 \neq x_2$, k must map x_1 to 0 and x_2 to 0, and thus must be the zero map. Thus, $\dim J = |H \setminus G| \dim J_x = |H \setminus G| \dim U = |G/H| \dim U$ $\square$

Problem 2.

Proof. $\forall h \in H$, $\forall x \in G$, $\forall F \in J_x$, $H \triangleleft G \implies \exists h' = xhx^{-1} \in H \implies h'x = xh$, $hF(x) = F(xh) = F(h'x) = h'F(x) \neq 0$ since h' is invertible and $F(x) \neq 0$. Thus J_x is H invariant.

$\forall X \in H \setminus G$, $\forall F \in J_x$, $\forall g, y \in G$. $gF(y) = F(yg)$. Only nonzero if $y \in Xg^{-1}$. Thus $gJ_X = J_{Xg^{-1}}$. Thus they are permuted the same way cosets are permuted.

Let $b_{X,i} \in J_X$ be the i th basis of J_X . $\forall y \in G$, $\forall a_{X,i} \in \mathbb{C} \ni \sum_{X \in H \setminus G} \sum_{i=1}^{\dim U} a_{X,i} b_{X,i}(y) = 0 \implies \sum_{i=1}^{\dim U} a_{yH,i} b_{yH,i}(y) = 0 \iff a_{yH,i} = 0$, thus all $a = 0$, thus J_{x_i} is linearly independent from J_{x_j} for $i \neq j$. Thus $J = \oplus J_x$ $\square$

Problem 3.

Proof. By 1, $J_X \cong U$ as a vectorspace. Moreover the isomorphism where $F \rightarrow F(e)$, $\forall h \in H$, $(hF)(e) = F(h) = hF(e)$, thus the isomorphism is an isomorphism of G reps. $\square$

Problem 4.

Proof. $\forall g, h \in H$, $tr(\rho(C_g(h))) = tr(\rho(ghg^{-1})) = tr(\rho(g)\rho(h)\rho(g)^{-1}) = tr(\rho(h))$, since tr is conjugacy invariant. By Frobenius's theorem, the reps are iso. For g outside of H , conjugation by g may move around the conjugacy classes, if so the traces would change, if not the traces would not change.

$\forall x \in G \forall h \in H$, $R_h(Hx) = Hxh = xHh = xH = Hx$, since $H \triangleleft G$.

$$\begin{array}{ccc} J_{Hx} & \xrightarrow{h} & J_{Hx} \\ \downarrow & & \downarrow \\ U & \xrightarrow{hxh^{-1}} & U \end{array}$$

$\forall x \in G$, $\forall h \in H$, the map $h : J_{Hx} \rightarrow J_{Hx}$, sends, $F \rightarrow hF$. The map from $J_{Hx} \rightarrow U$ sends $F \rightarrow F(x)$. So take a function F in the top left J_{Hx} , $F \rightarrow hF$, then we map down to U , so $hF \rightarrow (hF)(x) = F(xh) = F(xhx^{-1}x)$, since $H \triangleleft G$, $xhx^{-1} \in H$, and thus $F(xhx^{-1}x) = xhx^{-1}F(x)$. Thus $J_{Hx} \cong x(U)$ $\rho \rightarrow \rho \circ C_x$. $\square$

Problem 5.

Proof. Suppose $g \in G$, $\forall Hx \in H \setminus G$, $g \cdot J_{Hx} = J_{Hxg^{-1}}$, thus $Hx = Hxg^{-1} \implies \exists h \in H \ni hx = xg^{-1} \implies g^{-1} = x^{-1}hx \in H$ since $H \triangleleft G$ and $g \in H$.

$\forall h \in H$, by 4, h preserves J_x .

Thus on the diagonal of $g \notin H$, it would be 0s since J_X is not preserved $\forall X \in G/H$. Thus $tr(g) = 0$.

By 4, $\forall Hx \in H \setminus G$ $J_{Hx} \cong x(U)$ where $\rho \rightarrow \rho \circ C_x$, so $tr_{J_x}(h) = tr_U(xhx^{-1})$. Thus the trace of h is the sum over the blocks of the matrices which send J_x to J_x . Thus $tr_J(h) = \sum_{Hg \in H \setminus G} tr_{J_{Hg}}(h) = \sum_{g \in G} \frac{1}{|H|} tr_U(ghg^{-1})$, this is equal since $\forall x \in Hg$, $tr(xhx^{-1}) = tr(ghg^{-1})$.

Thus $tr_J(h) = \frac{1}{|H|} \sum_{g \in G} tr_U(ghg^{-1})$

Let $V|_H := res_H(V)$, be the restriction of the Representation of G , V , on H .

Let $J := ind_H^G(U)$.

$$\begin{aligned}
\langle Tr_J, Tr_V \rangle_G &= \frac{1}{|G|} \sum_{g \in G} tr_J(g) \overline{tr_V(g)} \\
&= \frac{1}{|G|} \sum_{h \in H} tr_J(h) \overline{tr_V(h)} \\
&= \frac{1}{|G|} \sum_{h \in H} \frac{1}{|H|} \sum_{g \in G} tr_U(ghg^{-1}) \overline{tr_V(h)} \\
&= \frac{1}{|G|} \frac{1}{|H|} \sum_{h \in H} \sum_{g \in G} tr_U(ghg^{-1}) \overline{tr_V(ghg^{-1})} \\
&= \frac{1}{|G|} \frac{1}{|H|} \sum_{h \in H} \sum_{g \in G} tr_U(h) \overline{tr_V(h)} \quad (\text{since elements of } H \text{ are simply reordered by conjugation}) \\
&= \frac{1}{|G|} \frac{1}{|H|} \sum_{h \in H} |G| tr_U(h) \overline{tr_V(h)} \\
&= \frac{1}{|H|} \sum_{h \in H} tr_U(h) \overline{tr_V(h)} \\
&= \langle Tr_U, tr_{V|_H} \rangle_H
\end{aligned}$$

□

Problem 6a.

Proof. Define $T := ind S : G \rightarrow GL_{2n} \forall x, y \in G$,

1. If $x, y \in H$, then $T(x)T(y) = \begin{bmatrix} S_x & 0 \\ 0 & S_{a^{-1}xa} \end{bmatrix} \begin{bmatrix} S_y & 0 \\ 0 & S_{a^{-1}ya} \end{bmatrix} = \begin{bmatrix} S_{xy} & 0 \\ 0 & S_{a^{-1}xa^{-1}a^{-1}ya} \end{bmatrix}$, because S is a homomorphism. $T(x)T(y) = \begin{bmatrix} S_{xy} & 0 \\ 0 & S_{a^{-1}xya} \end{bmatrix} = T(xy)$.
2. If $x \notin H$ and $y \in H$, then $T(x)T(y) = \begin{bmatrix} 0 & S_{xa} \\ S_{a^{-1}x} & 0 \end{bmatrix} \begin{bmatrix} S_y & 0 \\ 0 & S_{a^{-1}ya^{-1}} \end{bmatrix} = \begin{bmatrix} 0 & S_{xya} \\ S_{a^{-1}xy} & 0 \end{bmatrix} = T(xy)$
3. if $y \notin H$ and $x \in H$, then $T(x)T(y) = \begin{bmatrix} S_x & 0 \\ 0 & S_{a^{-1}xa} \end{bmatrix} \begin{bmatrix} 0 & S_{ya} \\ S_{a^{-1}y} & 0 \end{bmatrix} = \begin{bmatrix} 0 & S_{xya} \\ S_{a^{-1}xya} & 0 \end{bmatrix} = T(xy)$
4. If $x, y \notin H$, then $T(x)T(y) = \begin{bmatrix} 0 & S_{xa} \\ S_{a^{-1}x} & 0 \end{bmatrix} \begin{bmatrix} 0 & S_{ya} \\ S_{a^{-1}y} & 0 \end{bmatrix} = \begin{bmatrix} 0 & S_{xya} \\ S_{a^{-1}xya} & 0 \end{bmatrix} = T(xy)$

Thus T is a homomorphism and thus is a representation G . By 5,

$$tr(g) = \begin{cases} 0 & \text{if } g \in aH \\ \frac{1}{|H|} \sum_{x \in G} tr_S(xgx^{-1}) & \text{if } g \in H \end{cases} = \begin{cases} 0 & \text{if } g \in aH \\ tr_S(h) + tr_S(a^{-1}ga) & \text{if } g \in H \end{cases}$$

□

Problem 6b.

Proof. The traces of $T|_H$ and the traces of $S \oplus S'$ are equal as $\forall h \in H, tr_{T|_H}(h) = tr_T(h) = tr_S(h) + tr_S(a^{-1}ha) = tr_S(h) + tr_{S'}(h) = tr_{S \oplus S'}(h)$. By Frobenius's Theorem, $T|_H \cong S \oplus S'$. $\square$

Problem 6c.

Proof. Proved in 5 for general induced representations and general representations.

$$\begin{aligned}
\langle \chi_T, \chi_R \rangle &= \frac{1}{|G|} \sum_{g \in G} tr_T(g) \overline{tr_R(g)} \\
&= \frac{1}{|G|} \sum_{h \in H} (tr_S(h) \overline{tr_R(h)} + tr_S(a^{-1}ha) \overline{tr_R(h)}) \\
&= \frac{1}{|G|} \sum_{h \in H} (tr_S(h) \overline{tr_R(h)} + tr_S(a^{-1}ha) \overline{tr_R(a^{-1}ha)}) \\
&= \frac{1}{|G|} \left[\sum_{h \in H} (tr_S(h) \overline{tr_R(h)}) + \sum_{h \in H} (tr_S(a^{-1}ha) \overline{tr_R(a^{-1}ha)}) \right] \\
&= \frac{1}{|G|} \left[\sum_{h \in H} (tr_S(h) \overline{tr_R(h)}) + \sum_{h \in H} (tr_S(h) \overline{tr_R(h)}) \right] \\
&= \frac{2}{|G|} \sum_{h \in H} (tr_S(h) \overline{tr_R(h)}) \\
&= \frac{[G : H]}{|G|} \sum_{h \in H} (tr_S(h) \overline{tr_R(h)}) \\
&= \frac{1}{|H|} \sum_{h \in H} (tr_S(h) \overline{tr_R(h)}) \\
&= \langle \chi_S, \chi_{R|_H} \rangle
\end{aligned}$$

$\square$

Problem 6d.

Proof. Suppose S is an irrep of H .

Suppose $S \not\cong S'$, then $\langle \chi_T, \chi_T \rangle = \langle \chi_S, \chi_{T|_H} \rangle = \langle \chi_S, \chi_S + \chi_{S'} \rangle$, by part b. $\langle \chi_T, \chi_T \rangle = \langle \chi_S, \chi_S \rangle + \langle \chi_S, \chi_{S'} \rangle = 1 + 0 = 1$, by orthogonality relations. Thus by orthogonality, T is an irrep.

Suppose $S \cong S'$. Then by Frobenius reciprocity, $\langle \chi_T, \chi_T \rangle = \langle \chi_S, \chi_{T|_H} \rangle = 2$. Thus T must be reducible, and must be the direct sum of irreducibles of G . Furthermore for any irreducible $V \leq T$, $1 \leq \langle \chi_T, \chi_V \rangle = \langle \chi_S, \chi_{V|_H} \rangle$ which means $S \leq V \implies \dim V \geq \dim S$. Suppose T breaks down into n irreducibles $V_1, \dots, V_n$, the dimension of each irreducible is at least $\dim S$, thus $n \dim S \leq \sum_{i=1}^n \dim V_i = \dim T = 2 \dim S \implies n = 2$ and the dimension of each irreducible is $\dim S \implies \langle \chi_T, \chi_{V_i} \rangle = \langle \chi_S, \chi_{V_i|_H} \rangle = 1 \implies T$ must break down into the direct sum of 2 nonisomorphic irreducibles. $\square$

Problem 7.

Proof. $\mu_n \triangleleft D_n$ and $[D_n : \mu_n] = 2$. Since μ_n is abelian all reps are 1 dimensional, moreover all 1 reps are homomorphisms from $\mu_n \rightarrow \mu_n$. The group of 1 dimensional reps, $\{S_0, \dots, S_{n-1}\}$ is μ_n , by artin 4. $\forall k \neq 0$ and $k \neq \frac{n}{2}$ if n is even, $S_k \not\cong S'_k = S_k \circ C_y = S_{n-k} =: S_{k-1}$ since $S_k(x) = x^k$ and $S'_k(x) = x^{n-k} = x^{-k}$.

$T_k := \text{Ind}_{\mu_n}^{D_n}(S_k)$. By problem 6a, $\forall g \in \mu_n$ $T_k(g) = \begin{bmatrix} S_k(g) & 0 \\ 0 & S_{k-1}(g) \end{bmatrix} = \begin{bmatrix} g^k & 0 \\ 0 & g^k \end{bmatrix}$ and $\forall h \in C_n y$ $y C_n = C_n y$ since $\mu_n = C_n \triangleright D_n$, $h = x^a y$ for some $a \in \{0, \dots, n\}$ $T_k(h) = T_k(x^a y) = \begin{bmatrix} 0 & S_k(hy) \\ S_k(yh) & 0 \end{bmatrix} = \begin{bmatrix} 0 & S_k(x^a) \\ S_k(yx^a y) & 0 \end{bmatrix} = \begin{bmatrix} 0 & x^{ak} \\ x^{-ak} & 0 \end{bmatrix}$.

For $k \neq 0$ and $k \neq \frac{n}{2}$ if n is even, T_k is irreducible by problem 6d. However $T_k \cong T_{k-1}$, since $\chi_{T_k}(x) = x^k + x^{-k} = x^{-k} + x^k = \chi_{T_{k-1}}(x)$ and $\chi_{T_k}(y) = 0 = \chi_{T_{k-1}}(y)$. Thus set of nonisomorphic T_k when $k \neq 0$, is $\{T_k | k \in [1, \dots, \lfloor \frac{n-1}{2} \rfloor]\}$.

For $k = 0$, T_k is not irreducible but is the sum of 2 nonisomorphic irreducibles by 6b. $\chi_{T_k}(x) = 2$ and $\chi_{T_k}(y) = 0$. y is order 2 so if T is the sum 1 dimensional reps, $y \rightarrow \pm 1$ and $x \rightarrow 1$. Let t , trivial rep, send $x \rightarrow 1$ and $y \rightarrow 1$ and let s the rep send $x \rightarrow 1$ and $y \rightarrow -1$. Thus $T_k = s + t$. S and T are irreducible.

For n even and $k = \frac{n}{2}$, T_k is not irreducible but is the sum of 2 nonisomorphic irreducibles by 6b. $\chi_{T_{\frac{n}{2}}}(x) = -2$ and $\chi_{T_k}(y) = 0$. x has even order thus has to go to -1, since that is the only root of unity such that the sum would -2, and y has order 2 so must go to ± 1 , thus let u be the rep which sends to $x \rightarrow -1$ and $y \rightarrow 1$, and v be the rep which sends $x \rightarrow -1$ and $y \rightarrow -1$.

Thus the irreducibles are $\{T_k | k \in [1, \dots, \lfloor \frac{n-1}{2} \rfloor]\} \cup \{s, t\}$ if n is odd and if n is even, $\{T_k | k \in [1, \dots, \lfloor \frac{n-1}{2} \rfloor]\} \cup \{s, t, u, v\}$ $\square$

Problem 8.

Proof.

$$\begin{array}{ccc} U & \cong & J_H \xrightarrow{\alpha} W \\ & & \downarrow \beta \\ \bigoplus_{X \in H \backslash G} J_x & = & J \end{array}$$

$\forall X = Hx \in G/H$, $\forall g \in G$, $\forall F \in J_X \ni F \neq 0$, $\forall y \in G$, $gF = F \circ R_g$, thus $(gF)(y) = F(yg)$, $F(yg) \neq 0 \iff yg \in Hx \implies y = Hxg^{-1}$. Thus $xF \in J_{Hxg^{-1}}$.

Thus $\forall Hx \in G/H$, $\forall F \in J_{Hx}$, $\tilde{F} := xF \in J_H$, thus $F = x^{-1}\tilde{F}$. For everything to commute, β is forced to send $F \rightarrow x^{-1}\alpha(\tilde{F}) = x^{-1}\alpha(xF)$. The map is unique, because there is only 1 possibility on where to send F for everything to commute.

$\forall Hx \in G/H$, $\forall y \in Hx$, $y = hy$ and $Hx = Hy$ suppose $F \in J_{Hy}$, thus $F \rightarrow y^{-1}\alpha(yF) = (hx)^{-1}\alpha(hxF) = x^{-1}h^{-1}h^{-1}\alpha(xF) = x^{-1}\alpha(xF)$, since α is a map of h reps. Thus β is well defined and exists.

β exists and is the unique, map of G reps. $\square$

Problem 9.

Proof. Let $b_{X,i} \in J_X$ be the i th basis of J_X . $\forall y \in G, \forall a_{X,i} \in \mathbb{C} \ni \sum_{X \in H \setminus G} \sum_{i=1}^{\dim U} a_{X,i} b_{X,i}(y) = 0 \implies \sum_{i=1}^{\dim U} a_{yH,i} b_{yH,i}(y) = 0 \iff a_{yH,i} = 0$, thus all $a = 0$, thus J_{x_i} is linearly independent from J_{x_j} for $i \neq j$.

$\sum_{X \in H/G} \dim J_X = |G/H| \dim U = \dim J$ by 1. Thus $\bigoplus_{X \in H \setminus G} J_X \cong J$. Moreover $\forall F \in J$, for all $Hx \in G, \forall hx \in Hx, F(hx) = hF(x) = hF_{Hx}(x)$ for some $F_{Hx} \in J_X$, F can be broken down into a sum of functions $F_X \in J_X$ which act on each coset X independently and are zero everywhere else. Thus $\bigoplus J_x$ span J .

$\forall X \in H/G, J_X$ only contributes to the trace of g if $gJ_X = J_X$. $\forall F \in J_X \ni F, \forall y \in G, (gF)(y) = F(yg)$. $F(yg) \neq 0 \implies yg \in X \implies yg \in X \implies y \in Xg^{-1}$, thus $gF \in J_{Xg^{-1}}$. Thus $gJ_X = J_{Xg^{-1}}$. Thus J_X only contributes to the trace if $gJ_X = J_X \implies X = Xg^{-1}$.

We let $x_1, \dots, x_k, k = |H \setminus G|$ be representatives of the cosets – so the cosets are the Hx_i . We have a vector space iso $J_{Hx_i} \rightarrow U$ by $F \rightarrow F(x_i)$.

Suppose $gJ_{Hx} = J_{Hx} \implies Hxg^{-1} = Hx \implies Hxg^{-1}x^{-1} = H \implies xg^{-1}x^{-1} \in H$. Suppose $xg^{-1}x^{-1} \in H$, then $Hxg^{-1}x^{-1} = H \implies Hxg^{-1} = Hx \implies gJ_{Hx} = J_{Hx}$.

Suppose $gJ_{Hx} = J_{Hx} \implies J_{Hx} = g^{-1}J_{Hx}$

$\forall F \in J_{Hx}, g^{-1}F \in J_{Hx}$. Thus $(g^{-1}F)(x) = F(xg^{-1}) = F(xg^{-1}x^{-1}x)$, since $xgx^{-1} \in H$, $(g^{-1}F)(x) = F(xg^{-1}x^{-1}x) = xg^{-1}x^{-1}F(x)$. Thus $\rho \rightarrow \rho \circ C_x$, where ρ is a H rep. Thus $Tr_{J_{Hx}}(g) = Tr_U(xg^{-1}x^{-1})$.

Suppose $x' = hx, tr_U(x'g^{-1}(x')^{-1}) = tr_U(hxg^{-1}x^{-1}h^{-1}) = tr_U(h(xg^{-1}x^{-1})h^{-1}) = tr_U(xg^{-1}x^{-1})$.

Let $X = Hx$ and $gJ_X = J_X$, $tr_{J_X}(g) = tr_U(xg^{-1}x^{-1}) = \frac{1}{|H|} \sum_{x \in X} tr_U(xg^{-1}x^{-1})$, since the choice of representative element of the coset doesn't matter (trace is equal regardless), and $|X| = |H|$.

$$tr_{J_X}(g) = \begin{cases} 0 & \text{if } xg^{-1}x^{-1} \notin H \\ \frac{1}{|H|} \sum_{x \in X} tr_U(xg^{-1}x^{-1}) & \text{if } xg^{-1}x^{-1} \in H \end{cases}$$

Extend tr_U such that $tr_U(G \setminus U) = \{0\}$.

Thus $tr_{J_X}(g) = \frac{1}{|H|} \sum_{x \in X} tr_U(xg^{-1}x^{-1})$ because $xg^{-1}x^{-1} \notin H, tr_U(xg^{-1}x^{-1}) = 0$ regardless, thus 1st condition is implicitly taken care of.

Thus $\forall g \in G, tr_J(g) = \sum_{X \in G/H} tr_{J_X}(g) = \sum_{X \in G/H} \frac{1}{|H|} \sum_{x \in X} tr_U(xg^{-1}x^{-1}) = \frac{1}{|H|} \sum_{x \in G} tr_U(xg^{-1}x^{-1})$ $\square$